

interview questions

What is a list in Python? How do you create a list?

A list in Python is an ordered, mutable collection of items. You can create a list by placing comma-separated values inside square brackets.

What is a tuple in Python and how does it differ from a list?

A tuple is an ordered, immutable collection of items. Unlike lists, tuples cannot be changed after their creation. You create a tuple by placing comma-separated values inside parentheses.

What is a set in Python and how is it different from a list?

A set is an unordered collection of unique items. Unlike lists, sets do not allow duplicate elements. You can create a set by placing comma-separated values inside curly braces.

What is a dictionary in Python?

A dictionary is an unordered collection of key-value pairs. You can create a dictionary by placing comma-separated key-value pairs inside curly braces.

What is a string in Python? How do you create one?

A string is a sequence of characters enclosed in single, double, or triple quotes. For example:

```
my_string = 'hello'
my_string = "hello"
my_string = '''hello'''
```

What are the Boolean values in Python?

Boolean values in Python are True and False. They represent truth values and are often used in conditional statements.

Define immutable data types in Python.

Immutable data types are those that cannot be changed after creation. Examples include numbers, strings, and tuples.

What is the purpose of the print statement in Python?

The print statement in Python is used to display information on the screen. Its primary purpose is to output data, messages, or results to the console or standard output device. It allows programmers to communicate with users by presenting information in a readable format.

Why are comments important in Python code?

Comments are important in Python code for several reasons. They help to explain the purpose and functionality of code segments, making it easier for other programmers (and yourself) to understand the codebase. Comments also serve as documentation, providing context, explanations, or warnings about specific sections of code. Additionally, comments can help in debugging and troubleshooting by temporarily excluding or annotating problematic code

When would you use single-line comments versus multi-line comments?

- Single-line comments, denoted by the `#` symbol, are typically used for brief explanations or annotations on a single line of code.
- Multi-line comments, enclosed within triple quotes (`'''` or `"""`), are used for longer explanations spanning multiple lines. They are often used for function or module-level documentation.

What is the difference between int, float, and str data types in Python?

- **int:** Stands for integer. Represents whole numbers without any decimal point.
- **float:** Stands for floating-point number. Represents real numbers with a decimal point.
- **str:** Stands for string. Represents a sequence of characters, enclosed in single (`'`) or double (`"`) quotes.

Why should you avoid using Python keywords as variable names?

Python keywords are reserved words that have special meanings in Python and cannot be used as variable names. Using them as variable names can lead to confusion or errors in the code. It's essential to avoid using Python keywords as variable names to maintain code clarity and prevent unintended behavior.

Describe the rules for creating variables in Python.

- Variable names must start with a letter or the underscore character (`_`).
- Variable names cannot start with a number.
- Variable names can only contain alphanumeric characters and underscores (`A-z`, `0-9`, and `_`).
- Variable names are case-sensitive (`name`, `Name`, and `NAME` are different variables).
- Reserved words (keywords) cannot be used as variable names.

What is the purpose of the `id()` function in Python?

The

`id()` function in Python returns the identity (memory address) of an object. It can be used to determine if two variables reference the same object in memory or to track the changes in an object's identity during its lifecycle.

Explain the usage of the `type()` function. How would you determine the data type of a variable in Python?

The

`type()` function in Python returns the data type of an object. It is useful for dynamically determining the type of a variable or object at runtime.

You can determine the data type of a variable in Python using the

`type()` function. Simply pass the variable as an argument to `type()`, and it will return the data type of that variable. For example:

